Uncertainty

POLS 602
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Confidence Intervals

Standardized Estimator

If we draw multiple samples from a population and calculate an estimate X for each, then the sampling distribution:

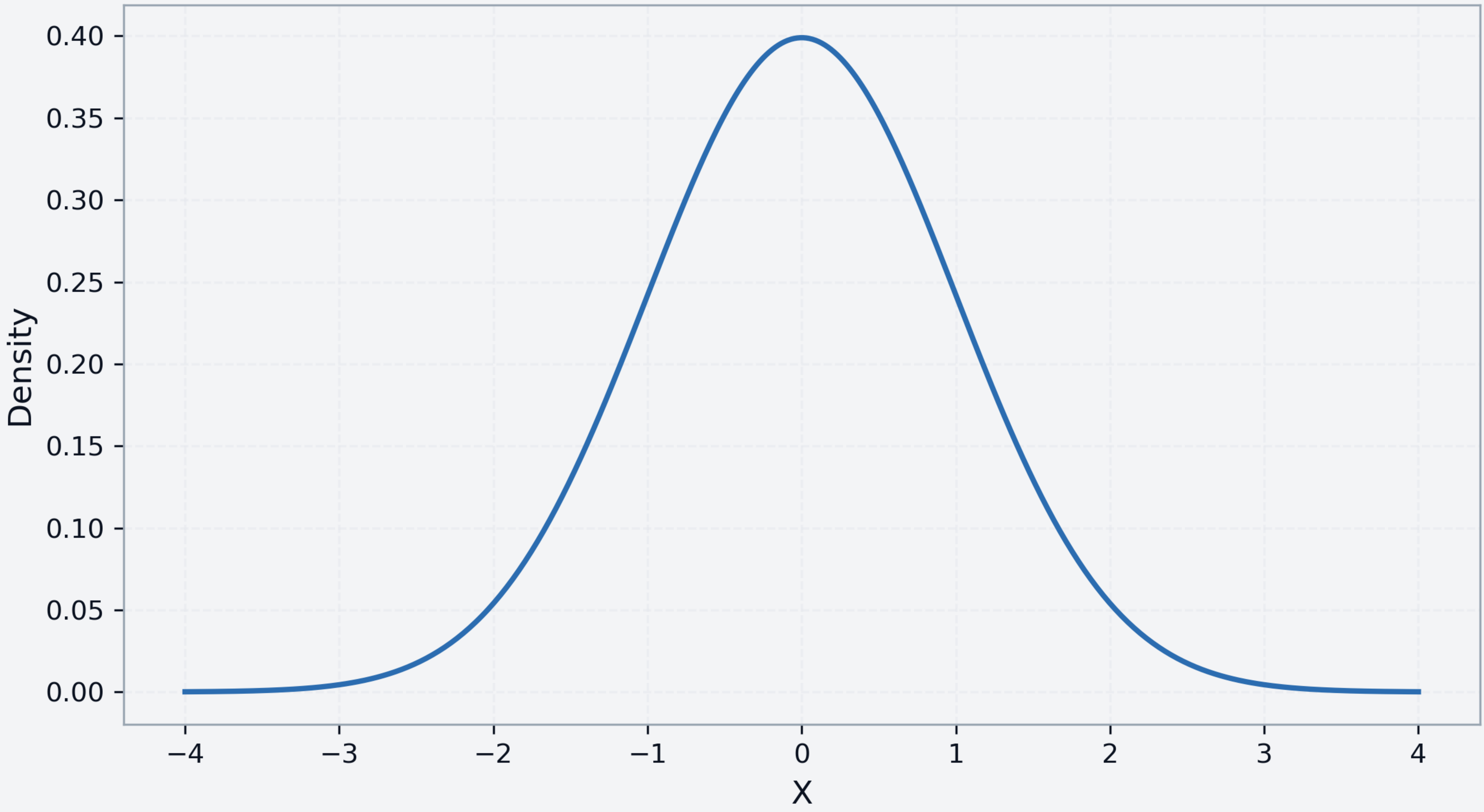
$$X \sim N(\text{true value}, (\text{standard error})^2)$$

The standardized estimate for X is:

$$\frac{X - \text{true value}}{\text{standard error}} \sim N(0,1)$$

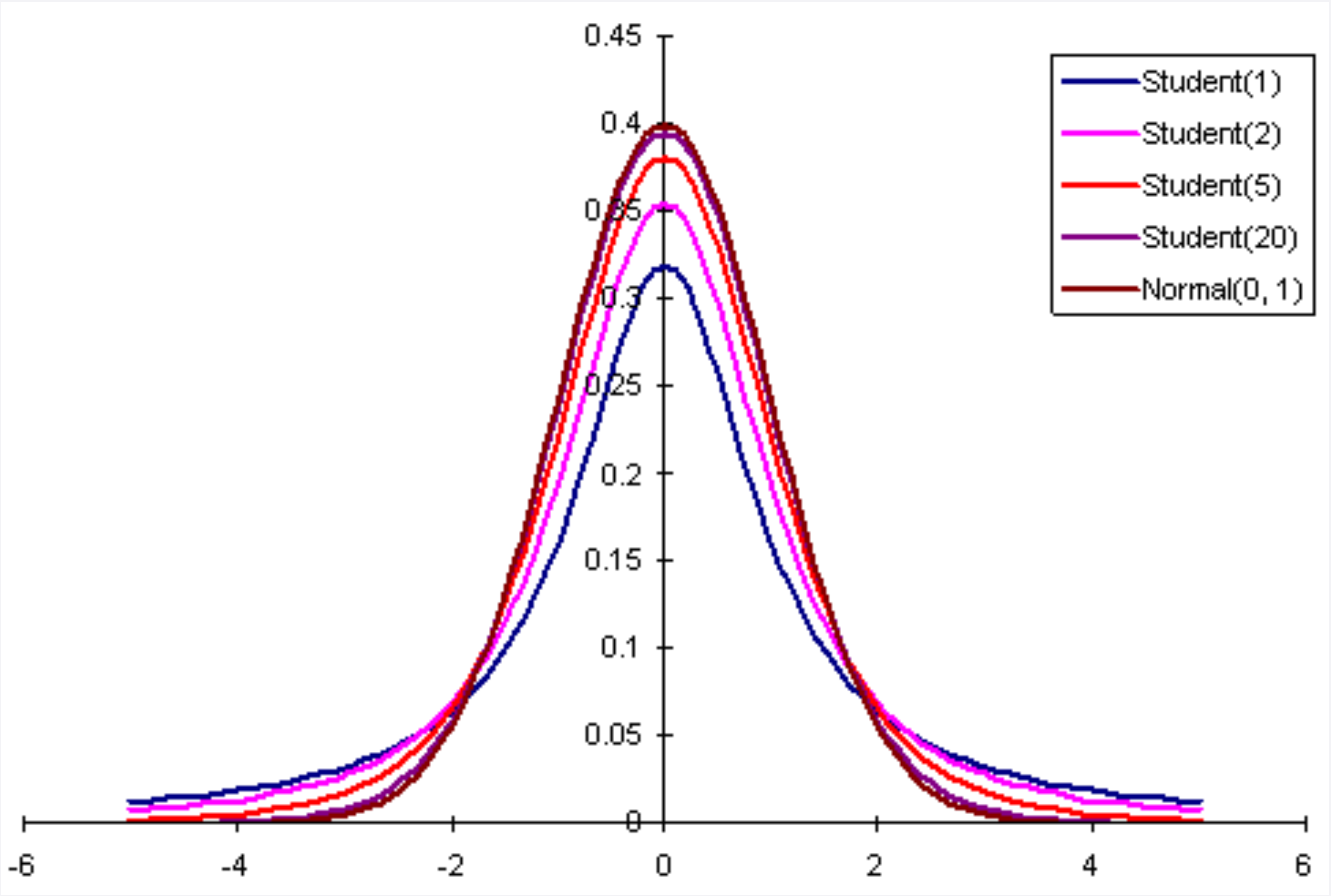
Z scores refresher

- Z denotes a standard normal random variable. That is: $X \sim N(0,1)$
- Thus, a 1 unit change in Z is a 1 standard deviation change
- Z scores are how far a random variable is from the mean in standard deviations
- 1.96 standard deviations from the mean covers approximately 95% of the standard normal distribution



t-statistic

- Z scores assume we have a large enough *n* so that the central limit theorem applies (i.e. the distribution is normal)
- We can relax this assumption by using the t-distribution
- The t-distribution is a zero mean distribution with the parameter v, which is called the degrees of freedom
- degrees of freedom = $n - k$ where n is the sample size and k is the number of parameters estimated
- $k = 1$ for the t-distribution, but you will see degrees of freedom elsewhere and k is not necessarily one in other contexts



t-statistic

$$\frac{\bar{X}_n - \mu}{\hat{\sigma}} \sim t_{n-1}$$

Confidence intervals

Confidence Interval: The range of values that is likely to include the true value of the parameter

$$P\left(-1.96 \leq \frac{X - \text{true value}}{\text{standard error}} \leq 1.96\right) \approx 0.95$$

$$P\left(X - 1.96 \times \text{standard error} \leq \text{true value} \leq X + 1.96 \times \text{standard error}\right) \approx 0.95$$

Hypothesis Testing

Key terms

Test statistic: A function of observed data that can be used to test the null hypothesis. Generally a z-statistic or a t-statistic.

p-value: The probability that an observed test statistic could have been observed by chance under an assumed distribution

Hypothesis testing procedure (NHST)

1. Specify a null hypothesis and an alternative hypothesis
2. Choose a test statistic and a level of test α
3. Derive the reference distribution — the sampling distribution of the test statistic under the null hypothesis
4. Compute the p-value for a one-sided or two-sided test
5. Reject the null hypothesis if the p-value is less than or equal to α

Hypothesis testing: t-test

$$P\left(X - 1.96 \times \text{standard error} \leq \text{true value} \leq X + 1.96 \times \text{standard error}\right) \approx 0.95$$

H_0 : true value $= \theta$

H_1 : true value $\neq \theta$

$$\frac{\bar{X}_n - \mu}{\hat{\sigma}} \sim N(0,1)$$

Hypothesis testing: t-test

$$P\left(X - 1.96 \times \text{standard error} \leq \text{true value} \leq X + 1.96 \times \text{standard error}\right) \approx 0.95$$

H_0 : true value $= \theta$

H_1 : true value $\neq \theta$

$$\frac{\bar{X}_n - \theta}{\hat{\sigma}} \sim N(0,1)$$

Hypothesis testing: t-test

$$P\left(X - 1.96 \times \text{standard error} \leq \text{true value} \leq X + 1.96 \times \text{standard error}\right) \approx 0.95$$

H_0 : true value $= 0$

H_1 : true value $\neq 0$

$$\frac{\bar{X}_n - 0}{\hat{\sigma}} \sim N(0,1)$$

Difference of means test

```{r}
# Generate data for control group and treatment group
control <- rnorm(30, mean = 50, sd = 10)
treatment <- rnorm(30, mean = 55, sd = 10)

# Two-tailed test
# H0: mean(control) = mean(treatment), difference of means = 0
# H1: mean(control) != mean(treatment)
two_tailed <- t.test(treatment, control, mu = 0,
                     alternative = "two.sided", conf.level = 0.95)
print(two_tailed)
```

```
	Welch Two Sample t-test

data:  treatment and control
t = 3.2645, df = 57.998, p-value = 0.001842
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
  2.89991 12.09392
sample estimates:
mean of x mean of y 
 54.46022  46.96331 
```

Difference of means test

```{r}
one_tailed <- t.test(treatment, control, alternative = "greater")
#one_tailed_less <- t.test(treatment, control, alternative = "less")

print(one_tailed)
```

```

        Welch Two Sample t-test

data:  treatment and control
t = 3.2645, df = 57.998, p-value = 0.0009212
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 3.658144      Inf
sample estimates:
mean of x mean of y 
 54.46022  46.96331 
```

One-sided vs. two-sided tests

- Two sided:
  - Detecting any deviation from the null hypothesis
  - No priors about the direction of the effect
  - More conservative, more widely used
  - Consider the consequences of being wrong in either direction
- One sided:
  - Theory predicts a specific direction
  - Evidence in the opposite direction is irrelevant
  - Increase statistical power when directional assumptions are justified

```
Call:
lm(formula = Y ~ X)

Residuals:
     Min       1Q   Median       3Q      Max 
-2.31841 -0.52163  0.04234  0.54490  2.42733 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  1.01801    0.02552   39.89   <2e-16 ***
X            2.43721    0.02036  119.73   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8069 on 998 degrees of freedom
Multiple R-squared:  0.9349,	Adjusted R-squared:  0.9348 
F-statistic: 1.433e+04 on 1 and 998 DF,  p-value: < 2.2e-16
```

Permutation test

A non-parametric test that generates, rather than assumes, a null distribution.

1. Compute the observed test statistic (e.g. a difference in means)
2. Randomly permute the data or labels
3. Re-compute the test statistic for the permuted data
4. Repeat the many times to build the empirical null distribution
5. Compare the observed statistic to the null distribution and compute the p-value

Permutation test

```{r}
treatment <- rnorm(30, mean = 55, sd = 10)
control   <- rnorm(30, mean = 50, sd = 10)

# Combine data
y <- c(treatment, control)
# Variable indicating true group
group <- c(rep("Treat", length(treatment)),
           rep("Control", length(control)))

# Observed difference in means
obs_diff <- mean(treatment) - mean(control)
print(obs_diff)
```

```
[1] 4.760618
```

Permutation test

```
```{r}
# Number of permutations
B <- 5000

# Vector to store permuted differences
perm_diffs <- numeric(B)

for (b in 1:B) {
  # Permute group labels
  permuted_group <- sample(group)

  # Compute difference in means under null (randomized) grouping
  perm_diffs[b] <- mean(y[permuted_group == "Treat"]) -
                   mean(y[permuted_group == "Control"])
}
```
```

Permutation test

```{r}
# How many of our permuted differences are great than
# or equal to observed differences?
p_value_two_sided <- mean(abs(perm_diffs) >= abs(obs_diff))

# One-sided p-values (optional)
p_value_greater <- mean(perm_diffs >= obs_diff)
p_value_less     <- mean(perm_diffs <= obs_diff)

cat("Two-sided p-value:", p_value_two_sided, "\n")
cat("One-sided p-value (Treat > Control):", p_value_greater, "\n")
cat("One-sided p-value (Treat < Control):", p_value_less, "\n")
```

```
Two-sided p-value: 0.0288
One-sided p-value (Treat > Control): 0.0136
One-sided p-value (Treat < Control): 0.9864
```

Permutation test

```{r}
# plot the distribution
hist(perm_diffs,
     breaks = 30,
     main = "Permutation Distribution of Difference in Means",
     xlab = "Permuted Differences")
abline(v = obs_diff, col = "red", lwd = 2)
```